

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
First Semester M.Tech. (EVT)(CBCS) Degree Examinations
 January 2023

22EVT133 - DIGITAL CONTROLLER

Duration: 3 Hours

Max. Marks:100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

		Unit – I			
		Marks	BT*	CO*	PO*
1.	a) Mention typical applications of dsPIC33 family microcontrollers.	6	L*1	1	2,5
	b) Explain the concept of interrupt nesting with an example.	6	L2	2	2,5
	c) Explain dataspace addressing with data memory map and associated block diagram.	8	L2	1	2,5
2.	a) Describe interrupt setup procedure for dsPIC33 interrupt controller.	8	L2	2	2,5
	b) List the features of interrupt controller.	6	L1	2	2,5
	c) What is modulo addressing mode? Mention how modulo addressing mode is realized in dsPIC33/PIC24 CPU family.	6	L2	1	2,5
3.	a) With block diagram explain how dsPIC33 series microcontrollers can be used for interleaved PFC application.	8	L3	1	2,5
	b) What is an interrupt vector table (IVT)? Comment on priority level of interrupts in IVT of dsPIC33 CPU.	6	L2	2	2,5
	c) Write a short note on Data Memory Space.	6	L1	1	2,5
		Unit – II			
4.	a) How to configure HRPWM module PWM generator clock? List the PWM modes that are available in HRPWM module.	8	L3	4	2,5
	b) Which peripheral modules can be used as trigger source for ADC? Mention triggering methods available in High-speed ADC module.	6	L2	3	2,5
	c) Write a note on Configurable Logic Cell (CLC).	6	L2	3	2,5
5.	a) Explain dual PWM mode of HRPWM operation.	7	L2	4	2,5
	b) Discuss the dedicated ADC core architecture with conversion sequence.	7	L3	3	2,5
	c) Which are the modes of operation supported by Timer1 module?	6	L1	4	2,5
6.	a) Explain 16-Bit Asynchronous External Counter Mode and give initialization steps.	8	L2	4	2,5
	b) Write a note on SAR ADC Cores.	6	L2	3	2,5
	c) Mention major applications of HRPWM module.	6	L1	4	2,5
		Unit – III			
7.	a) Referring to typical interconnection scheme, explain the I2C protocol procedures.	8	L3	5	2,5
	b) What are the key features of SPI module?	6	L1	5	2,5
	c) Write a note on CAN FD message filtering feature.	6	L2	5	2,5
8.	a) Explain the CAN FD module system block diagram.	7	L3	5	5,2
	b) What are the standards defined for I2C bus protocol?	7	L2	5	2,5
	c) List the modes of operation available in the SPI module.	6	L1	5	5,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
